

Milestone 1 Report

Project Title:

Climate Scope: Visualizing Global
Weather Trends and Extreme Events

Milestone 1: Data Preparation & Initial Analysis

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Introduction:

ClimateScope is a data visualization project aimed at analyzing global weather patterns and extreme climate events. The project uses the **Global Weather Repository dataset from Kaggle** to understand trends in weather parameters such as temperature, humidity, precipitation, and wind speed.

DV ClimateScope

The objective of this milestone is to prepare the dataset for analysis by performing **data acquisition, cleaning, preprocessing, and initial exploratory analysis.**

Dataset Description:

The dataset used in this project is the **Global Weather Repository dataset**, which contains worldwide weather observations updated daily.

Key Weather Variables

- Temperature
- Humidity
- Precipitation
- Wind Speed
- Location Information (Country / Region)
- Date and Time

These variables help analyze **seasonal trends, regional climate differences, and extreme weather events.**

Data Understanding and Exploration:

Initial exploration was performed to understand:

- Dataset structure
- Number of records
- Data types of columns
- Distribution of weather variables
- Presence of missing values

Key Observations

- Some columns contained missing values.
- Date columns required proper formatting.
- Weather variables showed different ranges across regions.

Initial Data Analysis:

Basic statistical analysis was conducted to understand the dataset.

Analysis Performed

- Summary statistics (mean, median, standard deviation)
- Distribution of weather variables
- Preliminary correlation analysis between variables

These insights will guide the **visualization design and deeper analysis in later milestones.**

Data Cleaning and Preprocessing:

Several preprocessing steps were applied to prepare the dataset for analysis.

Handling Missing Values

- Removed rows with excessive missing values.
- Filled missing values using appropriate statistical techniques where applicable.

Data Type Conversion

- Converted date columns to **datetime format**.
- Ensured numerical variables were correctly formatted.

Data Normalization

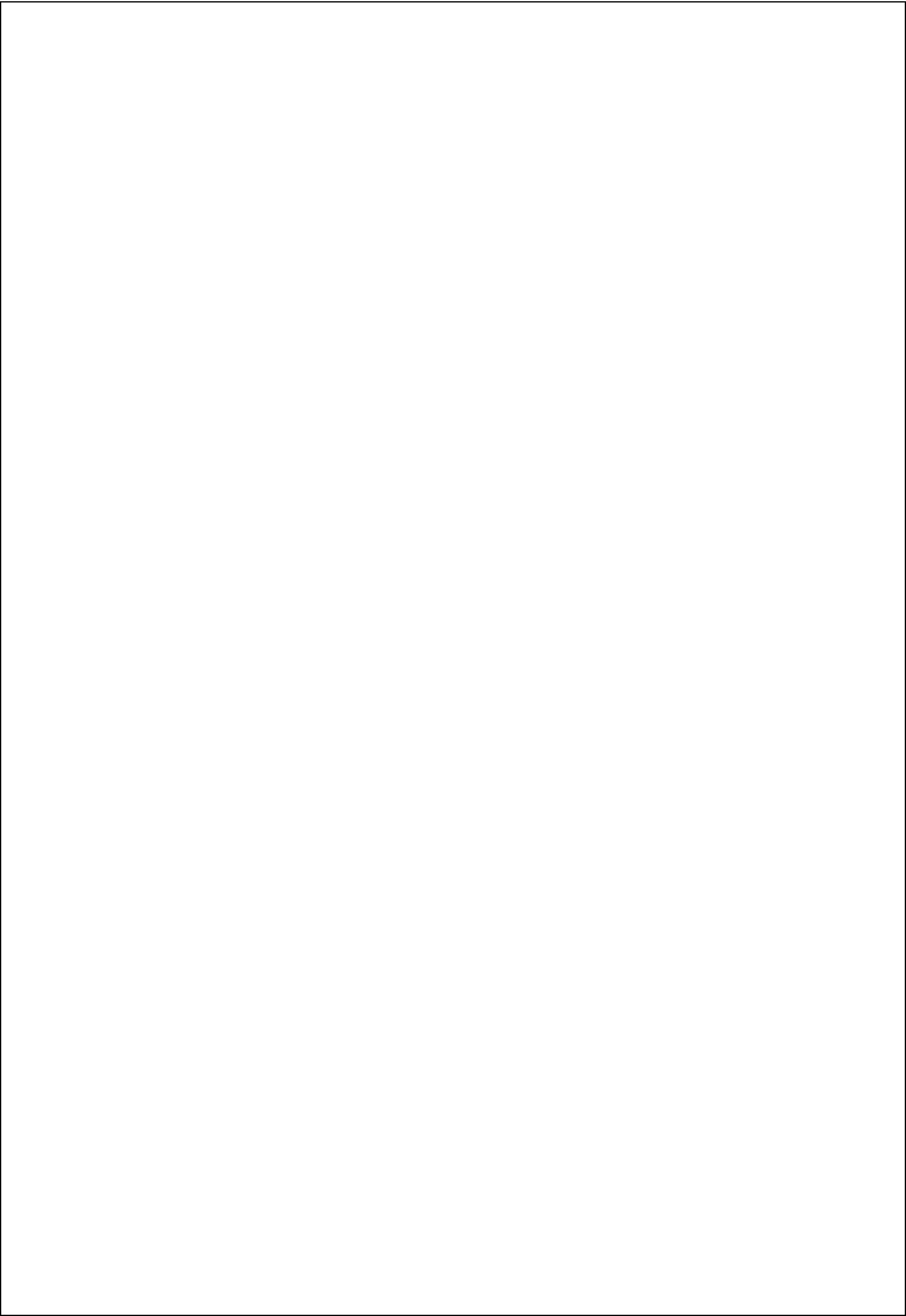
- Standardized units where necessary.
- Removed duplicate records.

Outcome of Milestone 1:

The following deliverables were successfully completed:

- Dataset downloaded and organized
- Dataset cleaned and pre-processed
- Initial exploration and analysis performed
- Dataset prepared for visualization and advanced analysis

The processed dataset is now ready for **statistical analysis and visualization development in Milestone 2.**



CLIMATESCOPE PROJECT

MILESTONE 1 : DATA PREPARATION AND INITIAL DATA ANALYSIS

In []:

IMPORT REQUIRED LIBRARIES

In [1]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from scipy.stats import pearsonr
```

DATA ACQUISITION – LOAD THE DATASET

In [2]:

```
data = pd.read_csv("../Dataset/GlobalWeatherRepository.csv")

data.head()
```

Out[2]:

	country	location_name	latitude	longitude	timezone	last_updated_epoch	last_updated
0	Afghanistan	Kabul	34.52	69.18	Asia/Kabul	1715849100	2024-05-16 13:15
1	Albania	Tirana	41.33	19.82	Europe/Tirane	1715849100	2024-05-16 10:45
2	Algeria	Algiers	36.76	3.05	Africa/Algiers	1715849100	2024-05-16 09:45
3	Andorra	Andorra La Vella	42.50	1.52	Europe/Andorra	1715849100	2024-05-16 10:45
4	Angola	Luanda	-8.84	13.23	Africa/Luanda	1715849100	2024-05-16 09:45

5 rows × 41 columns



DATASET STRUCTURE INSPECTION

In [3]:

```
data.columns
```

```
Out[3]: Index(['country', 'location_name', 'latitude', 'longitude', 'timezone',
              'last_updated_epoch', 'last_updated', 'temperature_celsius',
              'temperature_fahrenheit', 'condition_text', 'wind_mph', 'wind_kph',
              'wind_degree', 'wind_direction', 'pressure_mb', 'pressure_in',
              'precip_mm', 'precip_in', 'humidity', 'cloud', 'feels_like_celsius',
              'feels_like_fahrenheit', 'visibility_km', 'visibility_miles',
              'uv_index', 'gust_mph', 'gust_kph', 'air_quality_Carbon_Monoxide',
              'air_quality_Ozone', 'air_quality_Nitrogen_dioxide',
              'air_quality_Sulphur_dioxide', 'air_quality_PM2.5', 'air_quality_PM10',
              'air_quality_us-epa-index', 'air_quality_gb-defra-index', 'sunrise',
              'sunset', 'moonrise', 'moonset', 'moon_phase', 'moon_illumination'],
              dtype='object')
```

```
In [4]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 125111 entries, 0 to 125110
Data columns (total 41 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   country                               125111 non-null object
1   location_name                         125111 non-null object
2   latitude                             125111 non-null float64
3   longitude                             125111 non-null float64
4   timezone                             125111 non-null object
5   last_updated_epoch                   125111 non-null int64
6   last_updated                         125111 non-null object
7   temperature_celsius                  125111 non-null float64
8   temperature_fahrenheit                125111 non-null float64
9   condition_text                       125111 non-null object
10  wind_mph                             125111 non-null float64
11  wind_kph                             125111 non-null float64
12  wind_degree                           125111 non-null int64
13  wind_direction                       125111 non-null object
14  pressure_mb                           125111 non-null float64
15  pressure_in                           125111 non-null float64
16  precip_mm                             125111 non-null float64
17  precip_in                             125111 non-null float64
18  humidity                              125111 non-null int64
19  cloud                                125111 non-null int64
20  feels_like_celsius                   125111 non-null float64
21  feels_like_fahrenheit                 125111 non-null float64
22  visibility_km                         125111 non-null float64
23  visibility_miles                      125111 non-null float64
24  uv_index                             125111 non-null float64
25  gust_mph                             125111 non-null float64
26  gust_kph                             125111 non-null float64
27  air_quality_Carbon_Monoxide           125111 non-null float64
28  air_quality_Ozone                     125111 non-null float64
29  air_quality_Nitrogen_dioxide           125111 non-null float64
30  air_quality_Sulphur_dioxide            125111 non-null float64
31  air_quality_PM2.5                     125111 non-null float64
32  air_quality_PM10                      125111 non-null float64
33  air_quality_us-epa-index               125111 non-null int64
34  air_quality_gb-defra-index             125111 non-null int64
35  sunrise                               125111 non-null object
36  sunset                                125111 non-null object
37  moonrise                              125111 non-null object
38  moonset                               125111 non-null object
39  moon_phase                            125111 non-null object
40  moon_illumination                     125111 non-null int64
dtypes: float64(23), int64(7), object(11)
memory usage: 39.1+ MB
```

```
In [5]: data.describe()
```

Out[5]:

	latitude	longitude	last_updated_epoch	temperature_celsius	temperature_fahrenheit
count	125111.000000	125111.000000	1.251110e+05	125111.000000	125111.000000
mean	19.193775	21.992531	1.743688e+09	21.590124	70.863974
std	24.425628	65.794125	1.606001e+07	9.599724	17.279364
min	-41.300000	-175.200000	1.715849e+09	-29.800000	-21.600000
25%	3.750000	-6.836100	1.729848e+09	16.300000	61.300000
50%	17.250000	23.240000	1.743671e+09	24.100000	75.400000
75%	40.400000	50.580000	1.757578e+09	28.100000	82.600000
max	64.150000	179.220000	1.771484e+09	49.200000	120.600000

8 rows × 30 columns



In [6]:

data.shape

Out[6]: (125111, 41)

In [7]:

```
print("\n----- NUMERICAL FEATURES -----")
print(data.select_dtypes(include=['int64','float64']).columns)

print("\n----- CATEGORICAL FEATURES -----")
print(data.select_dtypes(include=['object']).columns)
```

```
----- NUMERICAL FEATURES -----
Index(['latitude', 'longitude', 'last_updated_epoch', 'temperature_celsius',
      'temperature_fahrenheit', 'wind_mph', 'wind_kph', 'wind_degree',
      'pressure_mb', 'pressure_in', 'precip_mm', 'precip_in', 'humidity',
      'cloud', 'feels_like_celsius', 'feels_like_fahrenheit', 'visibility_km',
      'visibility_miles', 'uv_index', 'gust_mph', 'gust_kph',
      'air_quality_Carbon_Monoxide', 'air_quality_Ozone',
      'air_quality_Nitrogen_dioxide', 'air_quality_Sulphur_dioxide',
      'air_quality_PM2.5', 'air_quality_PM10', 'air_quality_us-epa-index',
      'air_quality_gb-defra-index', 'moon_illumination'],
      dtype='object')
```

```
----- CATEGORICAL FEATURES -----
Index(['country', 'location_name', 'timezone', 'last_updated',
      'condition_text', 'wind_direction', 'sunrise', 'sunset', 'moonrise',
      'moonset', 'moon_phase'],
      dtype='object')
```

MISSING VALUE ANALYSIS

In [8]:

data.isnull().sum()

```
Out[8]: country 0
location_name 0
latitude 0
longitude 0
timezone 0
last_updated_epoch 0
last_updated 0
temperature_celsius 0
temperature_fahrenheit 0
condition_text 0
wind_mph 0
wind_kph 0
wind_degree 0
wind_direction 0
pressure_mb 0
pressure_in 0
precip_mm 0
precip_in 0
humidity 0
cloud 0
feels_like_celsius 0
feels_like_fahrenheit 0
visibility_km 0
visibility_miles 0
uv_index 0
gust_mph 0
gust_kph 0
air_quality_Carbon_Monoxide 0
air_quality_Ozone 0
air_quality_Nitrogen_dioxide 0
air_quality_Sulphur_dioxide 0
air_quality_PM2.5 0
air_quality_PM10 0
air_quality_us-epa-index 0
air_quality_gb-defra-index 0
sunrise 0
sunset 0
moonrise 0
moonset 0
moon_phase 0
moon_illumination 0
dtype: int64
```

Checking Duplicates

```
In [9]: data.duplicated().sum()
```

```
Out[9]: np.int64(0)
```

Convert Units

```
In [10]: # Convert Fahrenheit to Celsius (example column)
data["temp_c_from_f"] = (data["temperature_fahrenheit"] - 32) * 5/9

data[["temperature_fahrenheit", "temp_c_from_f"]].head()
```

Out[10]:

	temperature_fahrenheit	temp_c_from_f
0	79.8	26.555556
1	66.2	19.000000
2	73.4	23.000000
3	43.3	6.277778
4	78.8	26.000000

DATA NORMALIZATION

```
In [11]: from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
numeric_columns = data.select_dtypes(include=['int64', 'float64']).columns
data[numeric_columns] = scaler.fit_transform(data[numeric_columns])

print("\nNormalization completed")
```

Normalization completed

Aggregate Data (Daily → Monthly Average)

```
In [12]: data["last_updated"] = pd.to_datetime(data["last_updated"])

monthly_data = data.groupby(
    data["last_updated"].dt.to_period("M")
).mean(numeric_only=True)

monthly_data = monthly_data.reset_index()

monthly_data.head()
```

Out[12]:

	last_updated	latitude	longitude	last_updated_epoch	temperature_celsius	temperature_fahrenheit
0	2024-05	0.574965	0.554293	0.011268	0.695610	0.695327
1	2024-06	0.573443	0.555801	0.047542	0.712102	0.711832
2	2024-07	0.572694	0.556955	0.094316	0.716461	0.716193
3	2024-08	0.572853	0.556652	0.143049	0.716325	0.716056
4	2024-09	0.572446	0.556818	0.190377	0.695246	0.694982

5 rows × 32 columns



Save the Aggregated Data

```
In [13]: monthly_data.to_csv("../outputs/monthly_weather_data.csv", index=False)
```

Cleaned dataset exported

```
In [14]: data.to_csv("../outputs/cleaned_weather_data.csv", index=False)
```

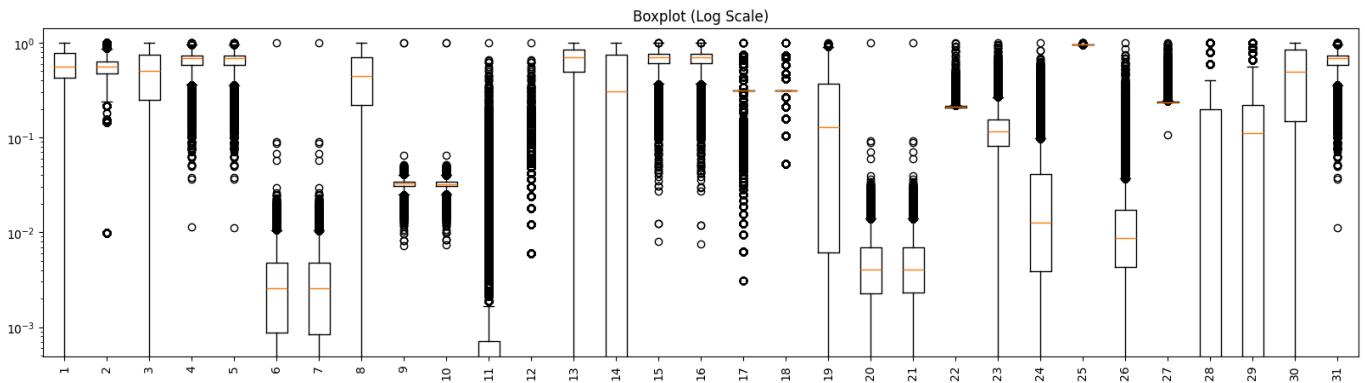
```
monthly_data.to_csv("../outputs/monthly_weather_data.csv", index=False)
```

```
print("\nCleaned dataset exported successfully")
```

Cleaned dataset exported successfully

Boxplot

```
In [15]: plt.figure(figsize=(20, 5))
plt.boxplot(data.select_dtypes(include='number'))
plt.yscale('log')
plt.xticks(rotation=90)
plt.title("Boxplot (Log Scale)")
plt.show()
```



In []: